

MODULE 2: Case Study - 1

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Objective

To create a Linux-based web server for XYZ Corporation, use an Amazon Machine Image (AMI) to reproduce the server, replicate it in another AWS Region, manage additional Amazon EBS volumes, resize and delete an EBS volume, and create a backup using an EBS snapshot.

Implementation Summary

Task	Requirement
1	Create Linux EC2 instance in US-East-1 and manage web servers using AMI
2	Replicate the instance in US-West-2 (Oregon)
3	Create two EBS volumes and attach them to the US-East-1 instance
4	Detach/delete one volume and extend the other
5	Back up the remaining EBS volume

1. Created Linux EC2 Web Server in US-East-1 (N. Virginia)

Step 1 – Region selected

The AWS console was used in US East (N. Virginia), identified as us-east-1.

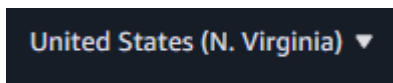


Figure 1. – Region Selection - US East (N. Virginia) – us-east-1

Step 2: Launched the EC2 instance

1. EC2 → Instances → Launch instances.
2. Name: **XYZ_server**
3. Selected Amazon Linux 2023 as the AMI
4. Selected t3.micro

5. Selected default VPC
6. Selected a key pair
7. In network settings; SSH, HTTPS and HTTP traffic are allowed.

Step 3: Web server configured

In EC2 User Data field, a startup script has been added that updated the system, installed httpd, enabled/started the service, and created /var/www/html/index.html with the Hello World / XYZ Corporation content.

Script:

```
#!/bin/bash
dnf update -y
dnf install -y httpd
systemctl enable httpd
systemctl start httpd

cat > /var/www/html/index.html <<'EOF'
<!DOCTYPE html>
<html>
<head><title>XYZ Corporation</title></head>
<body>
<h1>Hello World</h1>
<h2>XYZ Corporation Web Server</h2>
<p>Linux Web Server created using Amazon EC2 and AMI.</p>
</body>
</html>
EOF
```

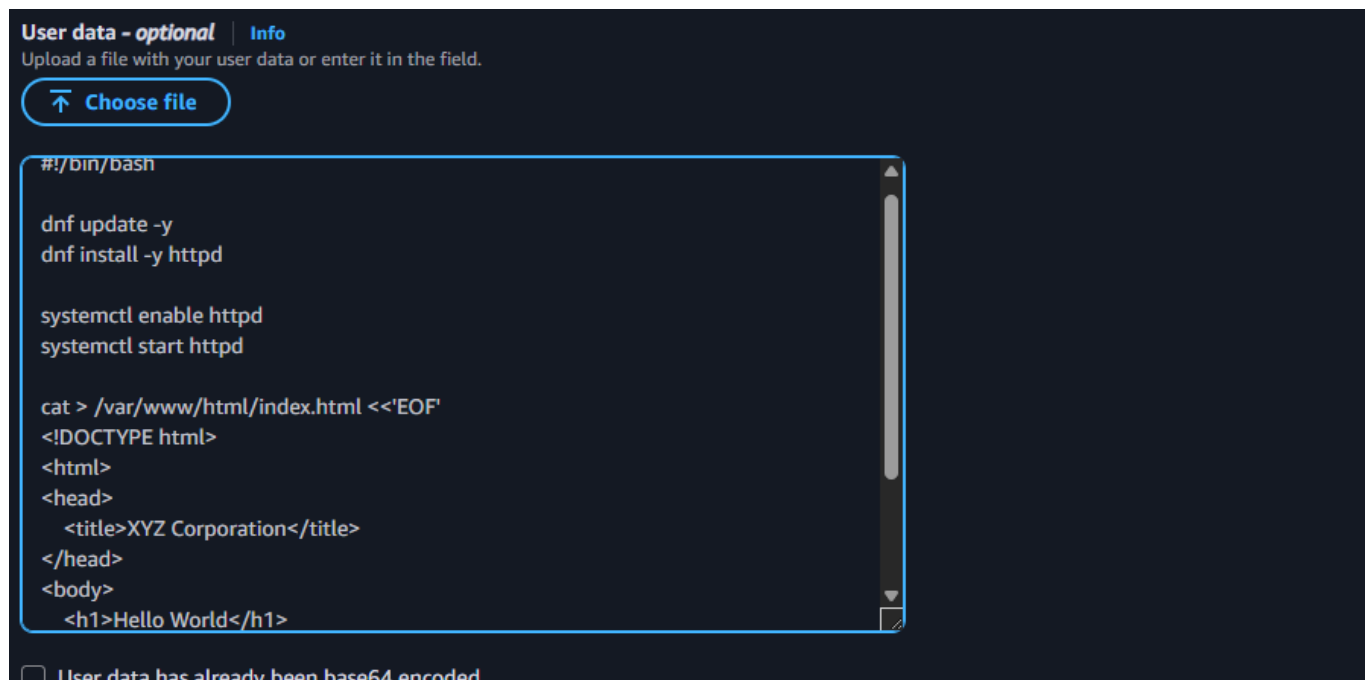


Figure 2. – User data added

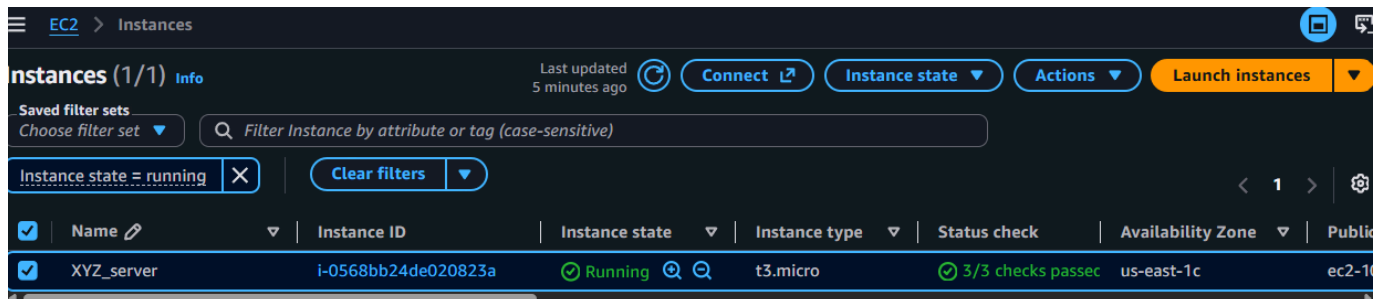


Figure 3. – Successfully created XYZ_server in US-East-1 (N. Virginia)

8. Launched the instance XYZ_server

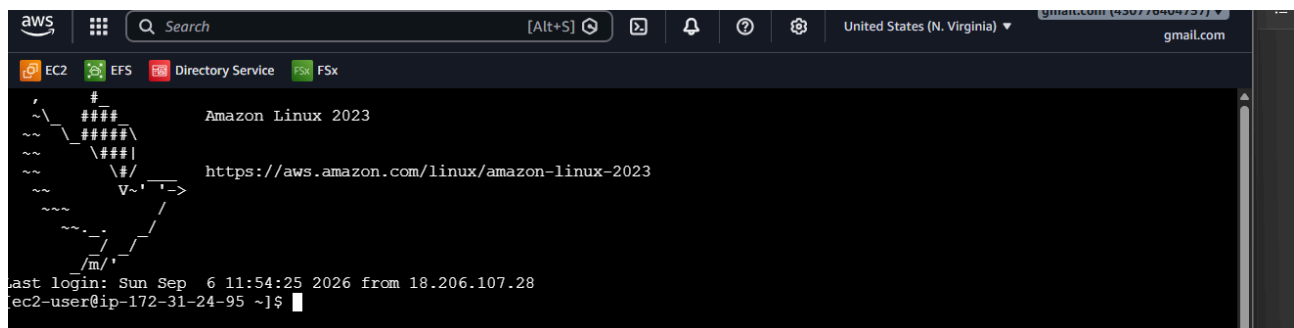


Figure 4. – Amazon Linux 2023 instance

Step 4: Test the web server

9. Using Public IP in a browser, the XYZ Corporation web page appears.

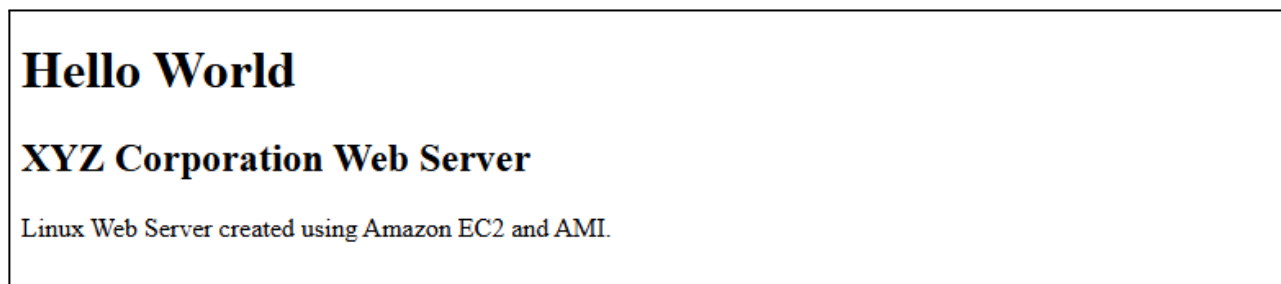


Figure 5. – XYZ corporation home page

2. AMI Created for the Web Server

The configured XYZ_server instance was captured as an Amazon Machine Image named XYZ-Web-Server-AMI.

- 10. EC2 → Instances and selected XYZ_server.
- 11. Actions → Image and templates → Create image.
- 12. image name: **XYZ-Web-Server-AMI**.

Image details

Instance ID

i-0568bb24de020823a

XYZ_server

Image name

XYZ-Web-Server-AMI

Maximum 127 characters. Can't be modified after creation.

Image description - optional

Image description

Maximum 255 characters

☒ Reboot instance

When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

Storage type	Device	Snapshot	Size	Volume type	IOPS	Throughput	Delete on termination	Encrypted
E...	/...	Create new sn...	8	EBS General P...	3000		☒ Enable	☐ Enable

Add volume

During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

☒ Tag image and snapshots together

Tag the image and the snapshots with the same tag.

☐ Tag image and snapshots separately

Tag the image and the snapshots with different tags.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

Create image

Figure 6. – creating XYZ-Web-Server-AMI

EC2 > AMIs

Amazon Machine Images (AMIs) (1/1) Info

Owned by me

Filter AMI by attribute or tag

Recycle Bin

EC2 Image Builder

Actions

Launch instance from AMI

	Name	AMI name	AMI ID	Source	Owner
☒	XYZ-Web-Server-AMI		ami-0260c74c2afa85c28	430776404757/XYZ-Web-Server-AMI	430776404757

Figure 7. – XYZ-Web-Server-AMI

3. Copied AMI to US-West-2 (Oregon)

An AMI is Region-specific, so copied the US-East-1 AMI to Oregon before launching the replica.

13. EC2 → AMIs.

Selected - XYZ-Web-Server-AMI.

14. Actions → Copy AMI.

15. Selected the destination to US West (Oregon) – us-west-2.

16. Name: **XYZ-Web-Server-AMI-Oregon** AMI.

The screenshot shows the 'Copy Amazon Machine Image (AMI)' console window. It includes fields for 'Original AMI ID' (ami-0260c74c2afa85c28), 'Region' (us-east-1 (N. Virginia)), 'AMI copy name' (XYZ-Web-Server-AMI), and 'AMI copy description' ([Copied ami-0260c74c2afa85c28 from us-east-1] XYZ-Web-Server-AMI). Under 'AMI destination', the 'AWS Region' option is selected. The 'Destination Region' dropdown is set to 'United States (Oregon)'.

Figure 8. – Copying AMI to US-West-2 (Oregon)

The screenshot shows the 'Amazon Machine Images (AMIs)' console. A table lists the copied AMI:

	Name	AMI name	AMI ID	Source	Owner
☒	XYZ-Web-Server-AMI-Oregon		ami-0c47ac42d83a32769	430776404757/XYZ-Web-Server-AMI-...	430776404757

Figure 9. – XYZ-Web-Server-AMI-Oregon

4. Instance Replicated to US-West-2 (Oregon)

17. Switched the Region to US West (Oregon).

18. EC2 → AMIs.

Selected the copied AMI - **XYZ-Web-Server-AMI-Oregon**

Launched instance from AMI.

19. Name: **XYZ-Web-Server-Oregon**

20. Selected a key pair, same network and security group configuration, user data, and launched instance.

21. Connected to instance and Checked the web page using the Oregon public IP.

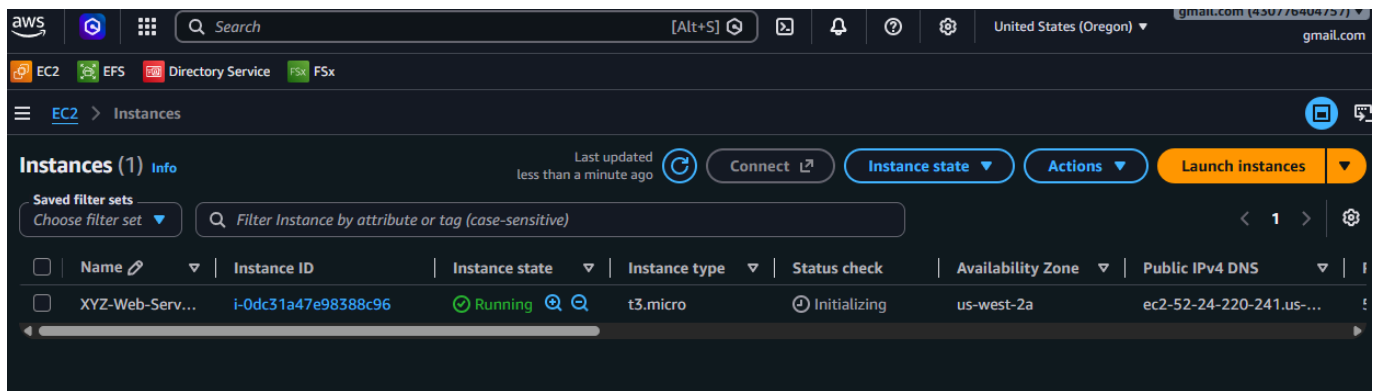


Figure 10. – Created XYZ-Web-Server-Oregon

5. Two EBS Volumes Created and Attached in US-East-1

Step 1: Availability Zone confirmed:

The EC2 instance was in us-east-1c. The EBS data volumes were therefore created in the same Availability Zone.

Step 2: Create EBS Volume 1

22. EC2 → Elastic Block Store → Volumes → Create volume.

23. Selected General Purpose SSD (gp3).

24. Size to 10 GiB.

25. Availability Zone to us-east-1c.

26. Selected Create volume.

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)
General Purpose SSD (gp3)

Size (GiB) [Info](#)
10
Min: 1 GiB, Max: 65536 GiB.

IOPS [Info](#)
3000
Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [Info](#)
125
Min: 125 MiB, Max: 2000 MiB, Baseline: 125 MiB/s.

Availability Zone [Info](#)
us-east-1c (us-east-1c)

Snapshot ID - optional [Info](#)
Don't create volume from a snapshot

Encryption [Info](#)
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☐ Encrypt this volume

Tags - optional [Info](#)
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
[Add new tag](#)
You can add up to 50 tags.

Figure 11 – Create 10 GiB gp3 EBS volume

Step 3: Create EBS Volume 2

27. A second 10 GiB gp2 EBS volume was created in the same Availability Zone (us-east-1c)

Step 4: Attached both volumes

28. Selected a 10 GiB volume → Actions → Attach volume → Selected XYZ_server.

Volumes (1/3) [Info](#)

Last updated less than a minute ago

[Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets
Choose filter set [▼](#)

	Name	Volume ID	Type	Size	IOPS
☒		vol-07762ea5c164882ae	gp3	10 GiB	3000
☐		vol-066474b9c683d25ca	gp2	10 GiB	100
☐		vol-0636add70f5ecc511	gp3	8 GiB	3000

Actions

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume**
- Detach volume
- Detach volume from instance

Source volume ID

-

-

-

Figure 12 – Attaching Volumes

29. Repeated the same for second volume
30. data-device names: /dev/sdd and /dev/sde
31. The following image shows that both volumes are attached to XYZ_server.

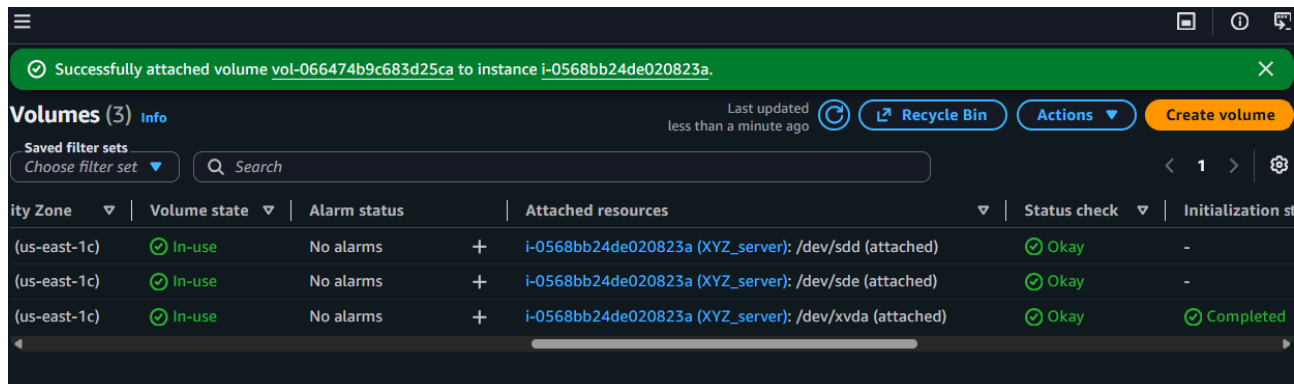


Figure 13 – Both EBS data volumes attached

Step 5: Verified from Linux

lsblk

```
/dev/nvme1n1
```

```
/dev/nvme2n1.
```

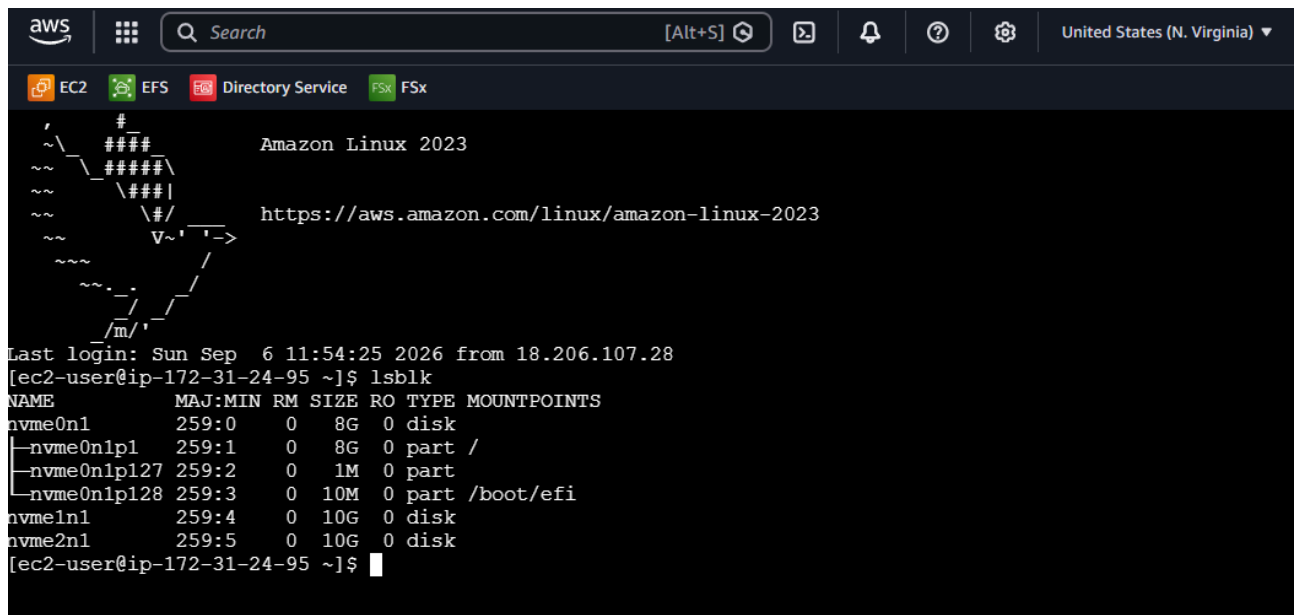


Figure 14 – lsblk confirmed two additional 10 GiB disks

6. One EBS Volume Detached and Deleted and Extended the Other

Step 1: Detached and deleted the Second volume

32. EC2 → Elastic Block Store → Volumes → Selected the 2nd 10GiB volume
Actions → Detach volume.
33. Selected the detached volume
Actions → Delete volume.

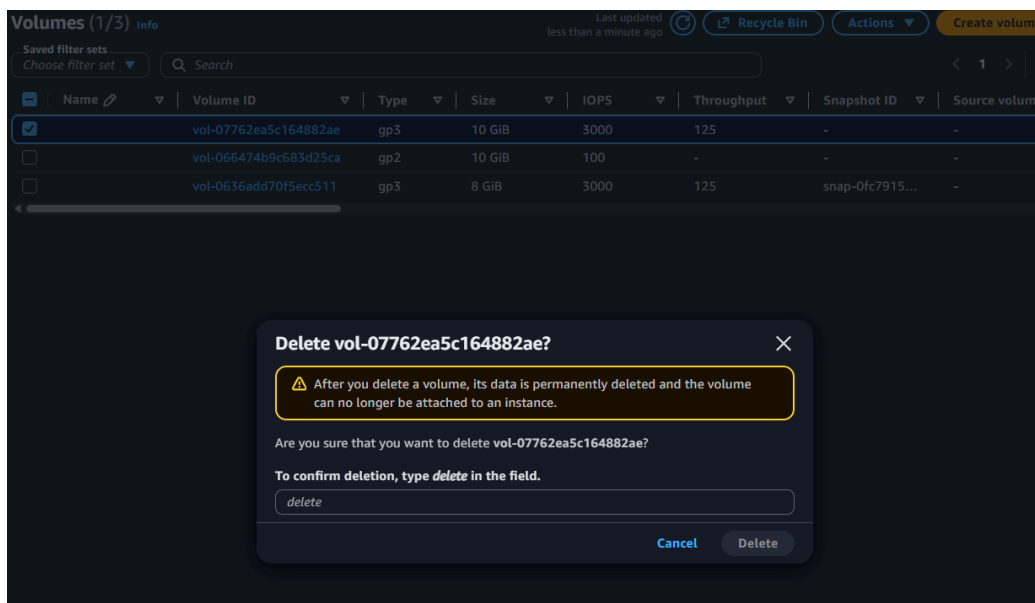


Figure 15 – Delete the detached EBS volume

Step 2: Extend the remaining EBS volume

34. Selected the remaining 10 GiB data volume.
Actions → Modify volume.
35. Changed Size from 10 GiB to 20 GiB.
→ Modify

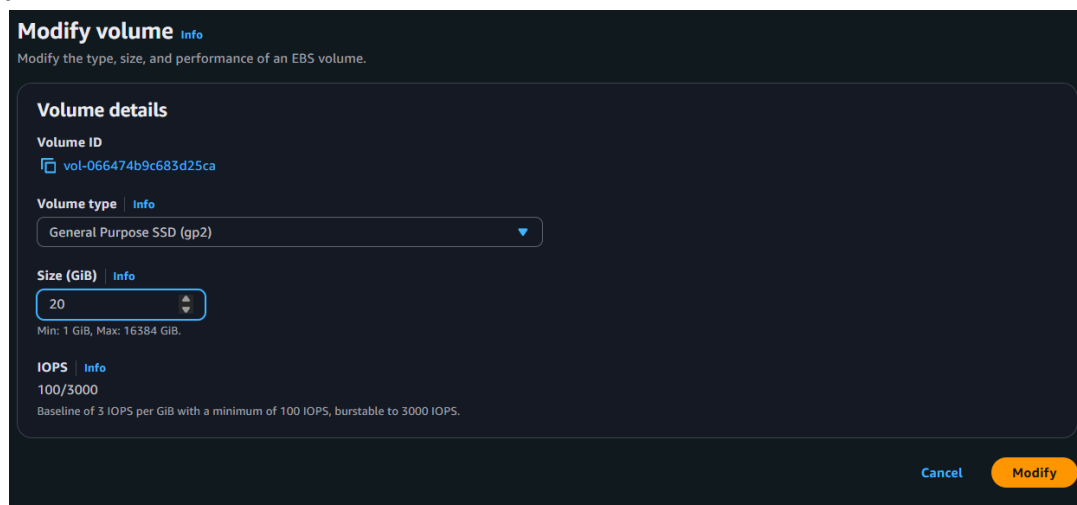


Figure 16 – Change EBS size to 20 GiB

Step 3: Verify the new size

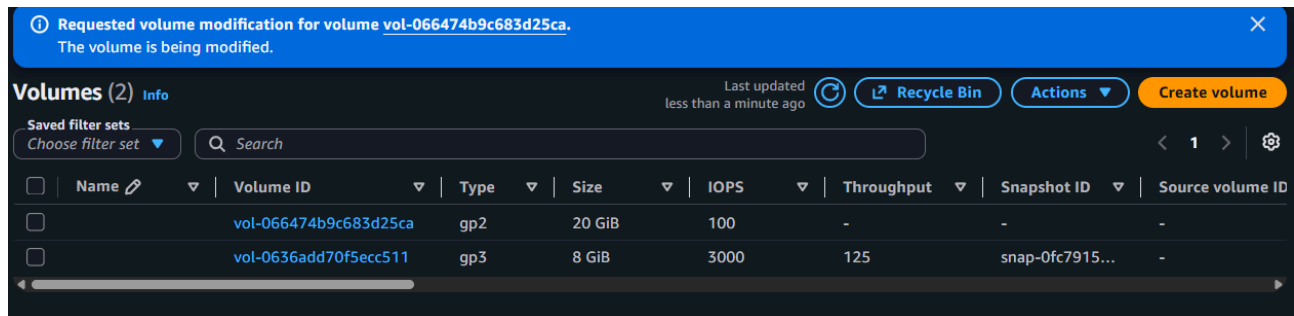


Figure 17 – AWS confirms the modification request

The subsequent `lsblk` output confirmed that `/dev/nvme2n1` increased from 10 GiB to 20 GiB.

```
Last login: Sun Sep  6 11:54:25 2026 from 18.206.107.28
[ec2-user@ip-172-31-24-95 ~]$ lsblk
NAME                MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1              259:0    0   8G  0 disk
├─nvme0n1p1          259:1    0   8G  0 part /
├─nvme0n1p127        259:2    0   1M  0 part
└─nvme0n1p128        259:3    0  10M  0 part /boot/efi
nvme1n1              259:4    0  10G  0 disk
nvme2n1              259:5    0  10G  0 disk
[ec2-user@ip-172-31-24-95 ~]$ lsblk
NAME                MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1              259:0    0   8G  0 disk
├─nvme0n1p1          259:1    0   8G  0 part /
├─nvme0n1p127        259:2    0   1M  0 part
└─nvme0n1p128        259:3    0  10M  0 part /boot/efi
nvme2n1              259:5    0  20G  0 disk
[ec2-user@ip-172-31-24-95 ~]$
```

Figure 18 – Linux verification after resize

7. Back Up the EBS Volume Using a Snapshot

Step 1 – Snapshot created:

36. EC2 → Elastic Block Store → Volumes
37. Selected the remaining 20 GiB EBS volume.
38. Actions → Create snapshot
39. description: to XYZ-Web-Server-EBS-Backup.
40. → Create snapshot

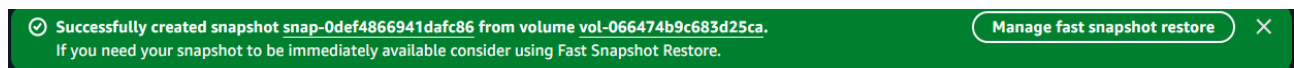


Figure 19 – Snapshot successfully created

Step 2 - Snapshot completion confirmed:

EC2 → Elastic Block Store → Snapshots.

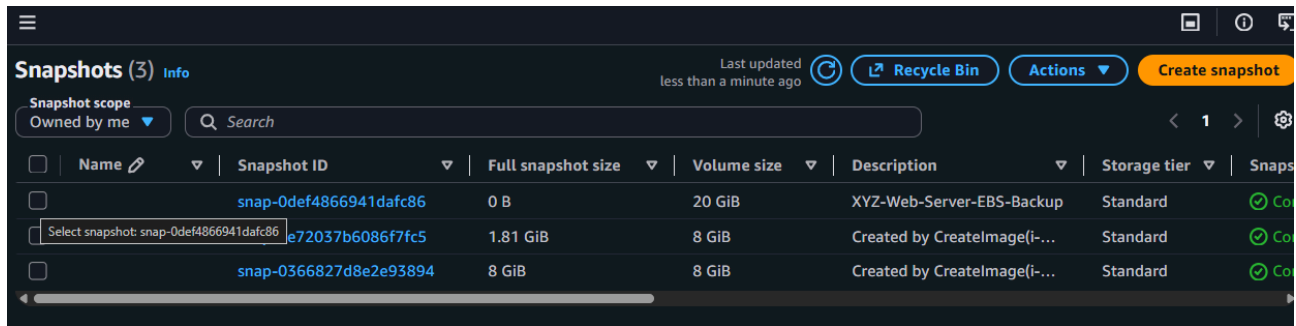


Figure 20 – Backup EBS snapshot created

Summary

Task	Completed Result	Evidence
1	Linux EC2 web server created in US-East-1 using Amazon Linux 2023 and configured with Apache.	Figures 1–5
2	Web-server AMI created and used for regional replication.	Figures 6–7
3	AMI copied and server replicated to US-West-2 (Oregon).	Figures 8–10
4	Two 10 GiB EBS volumes created in us-east-1c and attached to XYZ_server.	Figures 11–14
5	One data volume detached and permanently deleted.	Figure 15
6	Remaining data volume extended from 10 GiB to 20 GiB.	Figures 16–18
7	20 GiB EBS volume backed up with an EBS snapshot.	Figures 19–20